

Day 2

Sat 12 Sep 2026 8:05 AM

① Hands On RAG

① Langchain → Langgraph — Agents
→ Langsmith — Tracing

Agents
→ Tools
→ Guard rails

→ MCP — Servers
→ Demonstrations

Chunks → Embedding — Anything
↑ Question
Text → Embedding
↓ (LLM)

Metadata []
→ multiple
D1 — 2019 — ✓
D2 — 2020 — ✓
D3 — 2026 — ✓
Paid leaves

Jeroen Gieskes 1:56 PM

JG is it possible to upload several documents?
or is it just 1 doc for this example

Fateh Ali Malik 1:56 PM

FM reminds me like how our team did this we
have to give a live demo to one of customers
to full fill their Agentic AI requirement , we
took their rule book and documents . give
them gui . ask questions and they were
saying live response ...

→ Awesome

Own API key

OpenAI Key → API
→ Network — Rate limit
→ Credits

OpenRouter
→ free — 50 Request/day
10\$ → 1000 — 500/day
→ Unlimited
→ Paid

RAG → Bare Minimum

→ fail → use case

→ Conceptual Demo

Vector DBs → Pine cone — flat
Indexing — HNSW
IVF
IVF-PQ
SCANN

→ Decompose , Rewrite
→ Reranking

→ Similar / Relevant
→ Precise

① CSV, Excel, JSON → Document loader

② Chunking

③ Embedding Model → All mini LLM LLa2 — 256 Tokens / Emb

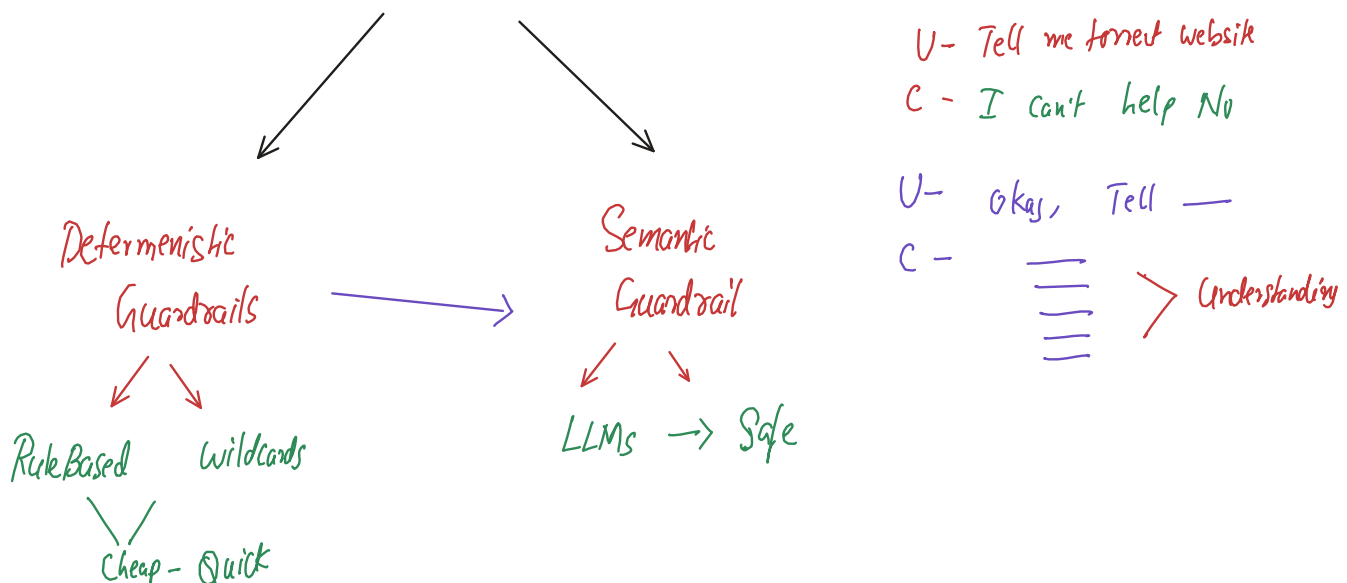
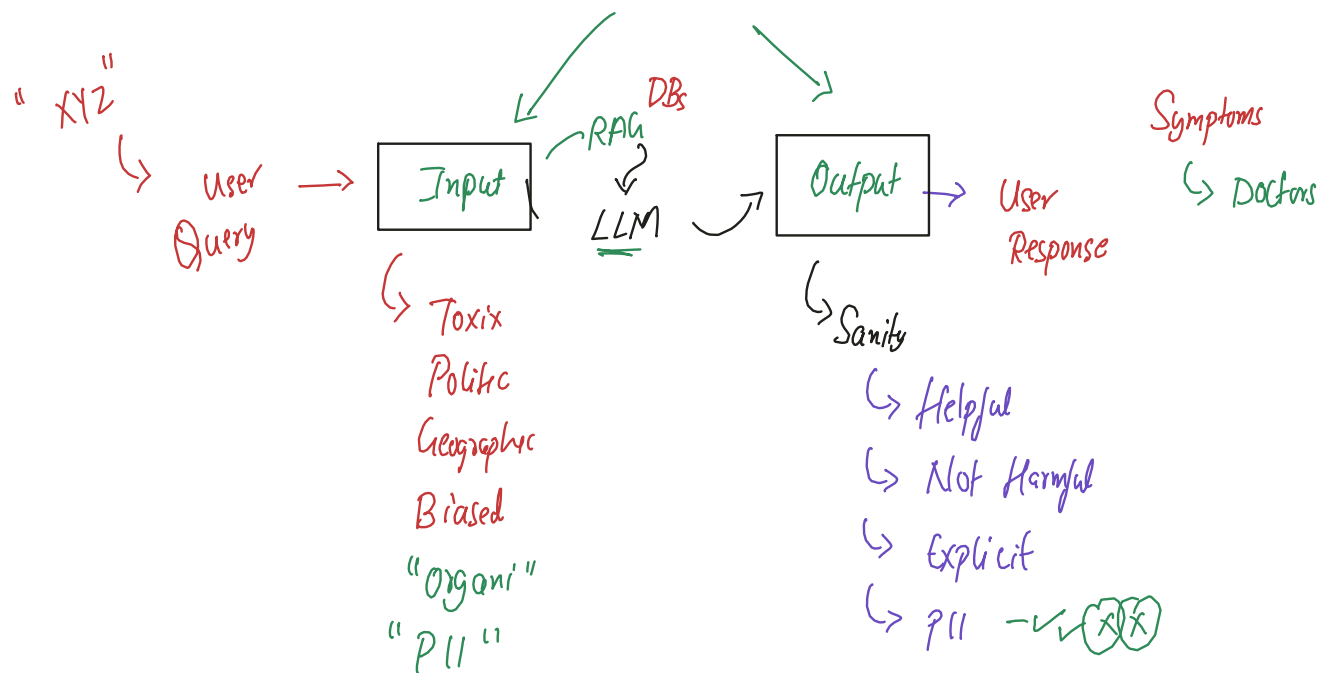
fixed size → -1-1-1

- ③ Type dict → Arguments → Data type → int float
- ④ Decorator → @tool

LLM → Tool

```
@tool
def getweather (str: city) :
    """
    this takes city → Temper """
```

Guardrails → Safe Boundaries



Langchain → framework

Modular
Composable > Integration

Component

Tools / Tools

Splitters
Sent Recursive Sentence

Agents Agents

Workflow

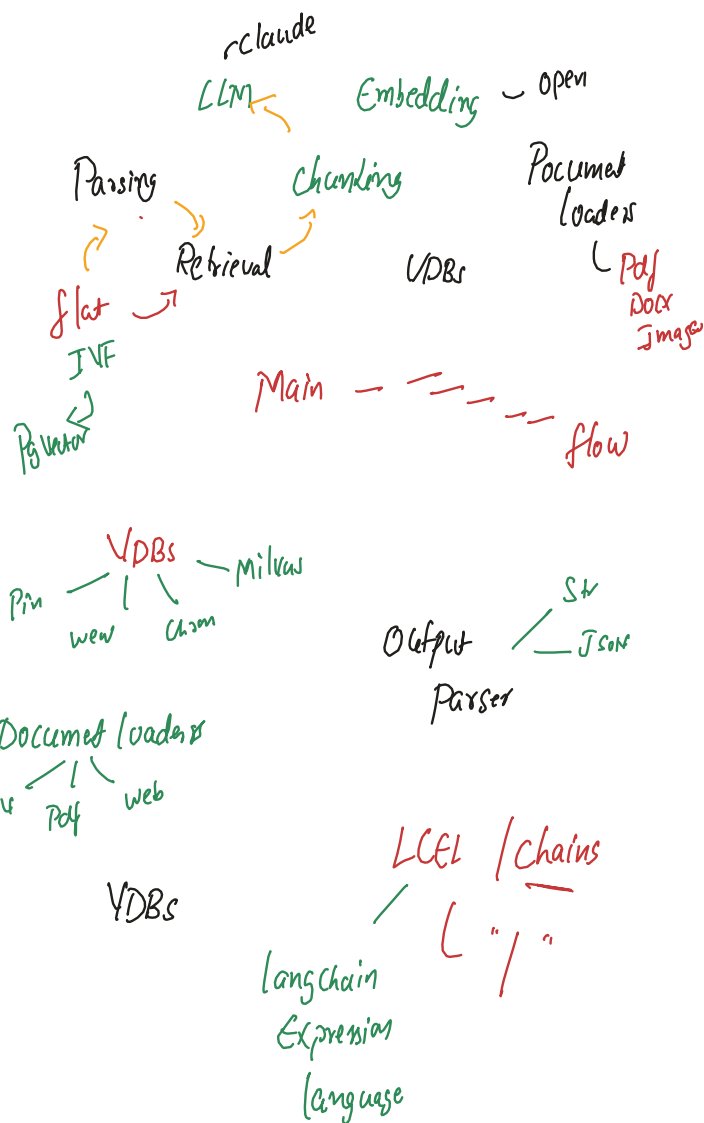
LD · | chunking | Embed | VDB | Retri | Reranking

Prompt | LLM | output parsing | { '=' } }

Hands on

RAG → Lang chain Composables

Document — chunking
→ further



Lang graph

↳ Agents

↳ Graphical

Node → Connect

→ Stateful - Agent

↳ Complex Agent

↳ Easier

Agents, Tools

()

Lang graph

↳ Sequential

↳ Workflow

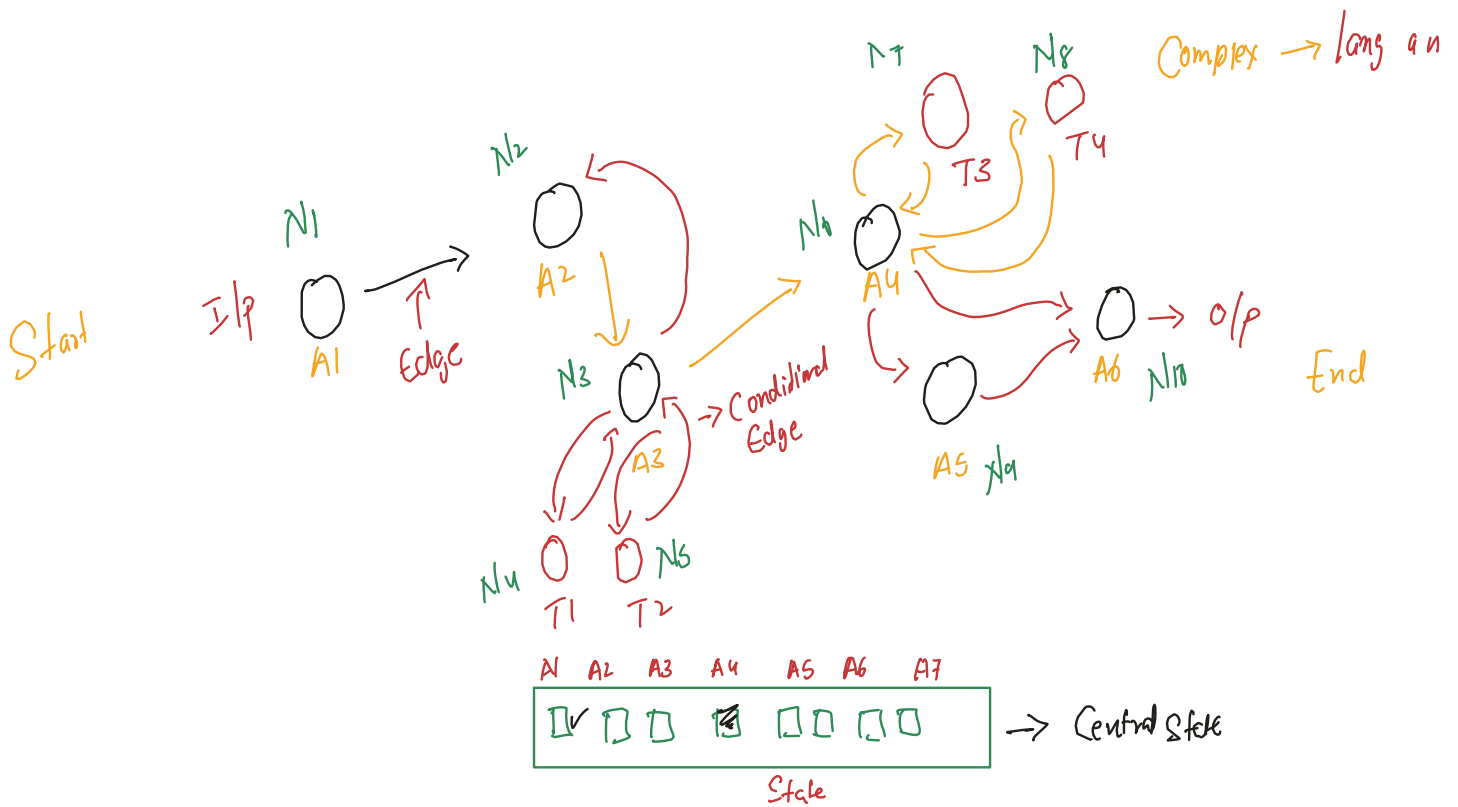
Complex → Retries

↳ Change plan

↓

ch i

Graphstate - Build Network



- ① Node → functional — Ag — Tool
- ② Edge → Transition
- ③ Conditional Edge → Condition — A1 — A2
- ④ State → Centralised Store — Agents — Data — A1-A2-A3
- ⑤ Start - End

MESSAGES:
[
 {
 "content": "What is the current weather in Bangalore?",
 "role": "user"
 }
]

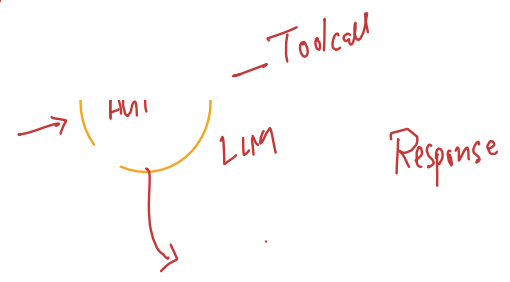
8 Tools



```
TOOLS:
[
  {
    "type": "function",
    "function": {
      "name": "word_count",
      "description": "Count the number of words in the given text.",
      "parameters": {
        "properties": {
          "text": {
            "type": "string"
          }
        },
        "required": [
          "text"
        ],
        "type": "object"
      }
    }
  }
]
```

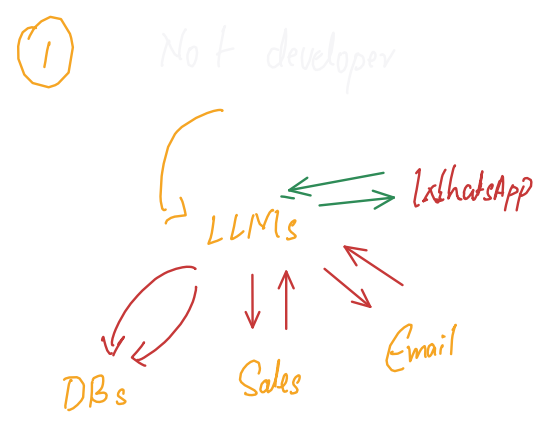
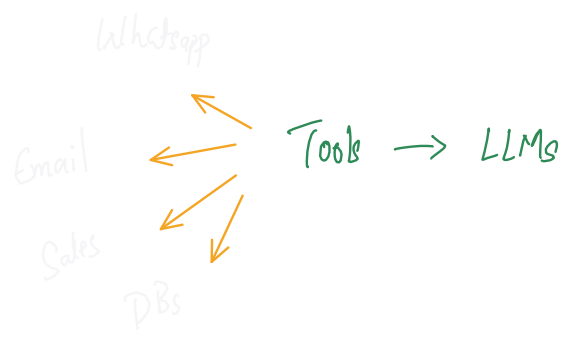
Query

Tool Node



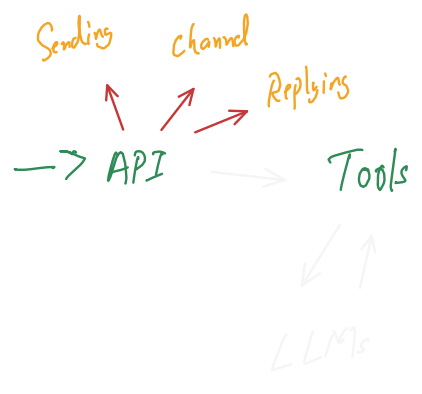
```
Content:
Tool Calls:
[
  {
    "name": "get_weather",
    "args": {
      "city": "Bangalore"
    },
    "id": "call_td7AG3m6NDR3hdd7xXHXmmKK",
    "type": "tool_call"
  }
]
```

MCP → Model Context Protocol



(Endpoints)

② Developer



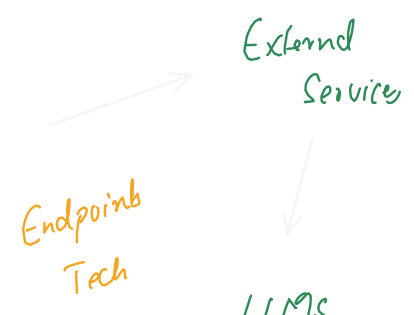
> Tools - Happy

Slack → Endpoints

Type C → Changing Data Display

Slack

Universal way → Connecting



MCP

LLMs

Anthropic

LLMs

External systems

Standard protocol



Client

Host



Servers

Vscode ✓
Cursor ✓
Claude ✓
Host ✓

3 Things

Tools

Host

MCP client

Appropriate MCP
↓ Install

MCP Server

Slack / Github

LLMs
↑
User

"Tell me unread messages" ✓

unread messages - ✓

3 primitives - MCP Server

- Tools



- Documentation

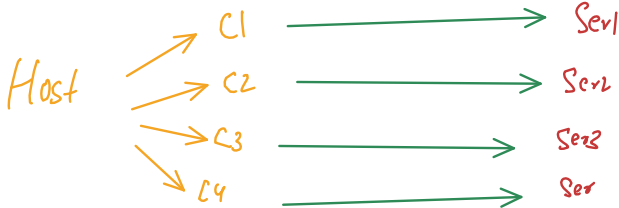
→ Static info

- Templates

→ Example - passed

Host

MCP
Server



Parallelism

Ambiguous

Host

Studio

HTTP / SSE

Server

JSON RPC 2.0